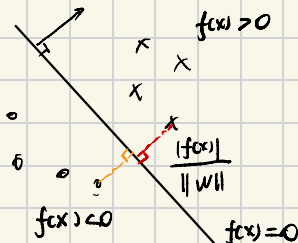


## Lecture 9 Support Vector Machine

### linear classifier

$$f(x) = w^T x + b$$

$$y \in \begin{cases} +1 & f(x) > 0 \\ -1 & f(x) < 0 \end{cases}$$



Distance from point  $x$   
to the boundary  $\frac{|f(x)|}{\|w\|}$

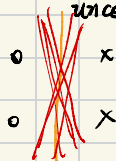
if the data is linearly separable, distance from the closest point to the boundary: "margin distance"

$$\gamma = \min_i \frac{|f(x_i)|}{\|w\|} = \min_i \frac{|w^T x_i + b|}{\|w\|} = \min_i \frac{y_i (w^T x_i + b)}{\|w\|}$$

Note: the scale will not affect  $\gamma$

Idea of SVM: maximize the better separate the boundary and the closest point

- ① perceptron: margins determine learning complexity
- ② trying points are random  $\rightarrow$  larger margin handles more uncertainty



- ③  $w$  is an uncertain estimate  $\rightarrow$  all more wiggle room w/o error

### Foundation of SVM

fix the nominator:  $\min_i |w^T x_i + b| = 1$

$$\text{objective: } \max_w \frac{1}{\|w\|}$$

Optimization prob

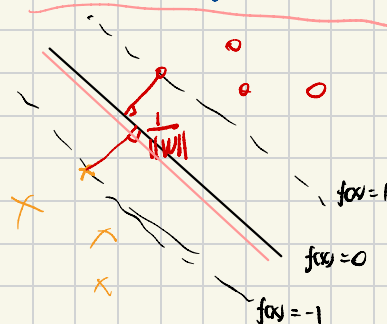
$$\arg \max_{w, b} \gamma \quad \text{s.t.} \quad \min_i |w^T x_i + b| = 1$$

$$\arg \min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad \min_i |w^T x_i + b| = 1$$

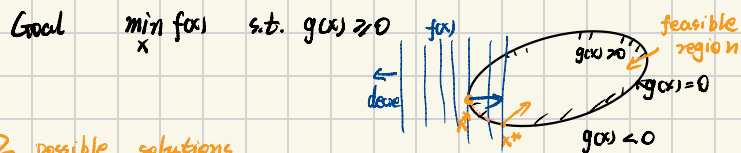
(at the optimum,  $w$  will skirt at least one point  $x_i$  will satisfy  $|w^T x_i + b| = 1$ )

$$\arg \min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y_i (w^T x_i + b) \geq 1 \quad \forall i$$

SVM problem



by the symmetry at least one point on each side.



2 possible solutions

1)  $x^*$  on the boundary  $g(x^*) = 0$  (active equality)

$$\text{at the minimum: } \nabla f(x) = \lambda \nabla g(x) \quad \lambda \geq 0$$

Both gradients point into the feasible region, and cannot decrease  $f$  without leaving the boundary

2)  $x^*$  is in the feasible region  $g(x^*) > 0$  (inactive equality)

$$\min \nabla f(x^*) = 0 \Rightarrow \lambda = 0 \text{ from ①}$$

Combining

$$L(x, \lambda) = f(x) - \lambda g(x)$$

$$\nabla f(x) - \lambda \nabla g(x) = 0$$

At the optima

$$\begin{cases} g(x) \geq 0 \\ \lambda \geq 0 \\ \lambda \cdot g(x) = 0 \end{cases}$$

} KKT condition

Karush-Kuhn-Tucker

Suppose we have optimal  $\lambda^*$ , minimize  $L(x, \lambda^*)$ 

$$L^* = \min_x L(x, \lambda^*) = \min_x [f(x) - \lambda^* g(x)] = \min_x f(x) = f(x^*)$$

at  $x^*, \lambda^* g(x^*) = 0$

$$\text{Define } q(\lambda) = \min_x L(x, \lambda) = \min_x [f(x) - \lambda g(x)]$$

for every  $\lambda$ , find min w.r.t.  $x$

$$\text{Note } \lambda \geq 0, g(x) \geq 0 \Rightarrow \lambda g(x) \geq 0$$

$$q(\lambda) \leq \min_x f(x) = f(x^*)$$

maximizing  $q(\lambda)$  could obtain  $f(x^*) = f^*$ Define Dual problem:  $q^* = \max_{\lambda \geq 0} q(\lambda)$ weak duality theorem:  $q^* \leq f(x^*)$  duality gap

strong duality theorem:

if ①  $f(x)$  is convex, ② feasible region is convex

③ feasible region is not degenerate

Then  $q^* = f(x^*)$  i.e. solve dual is equivalent to solve the original (primal)

# Lecture 9 Support Vector Machine

SVM problem:  $\arg \min_{w, b} \frac{1}{2} \|w\|^2$ , s.t.  $y_i (w^T x_i + b) \geq 1$

let  $d_i \geq 0$  be the lagrange multiplier for the  $i$ -th constraint

$$\text{Define: } L(w, b, d) = \underbrace{\frac{1}{2} \|w\|^2}_f - \underbrace{\sum_{i=1}^N d_i}_{\lambda} \underbrace{[y_i (w^T x_i + b) - 1]}_g$$

$$\text{Dual function } \approx L(d) = \min_{w, b} L(w, b, d)$$

$$\frac{\partial}{\partial w} L = w - \sum_i d_i y_i x_i = 0 \Rightarrow w^* = \sum_{i=1}^N d_i y_i x_i$$

$$\frac{\partial}{\partial b} L = -\sum_i d_i y_i = 0 \Rightarrow \sum_i d_i y_i = 0$$

tutorial

$$L(d) = \sum_{i=1}^N d_i - \frac{1}{2} \sum_i \sum_j d_i d_j y_i y_j x_i^T x_j$$

$$\text{SVM Dual problem } \max_d L(d) \text{ s.t. } d_i \geq 0, \sum d_i y_i = 0$$

$$\text{Given } d^* \Rightarrow w^* = \sum_{i=1}^N d_i^* y_i x_i$$

Recall KKT

$$\textcircled{1} g(x) = 0 \text{ active: } d_i > 0,$$

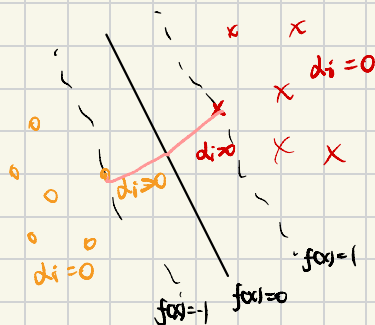
$$y_i (w^T x_i + b) = 1$$

some  $x_i$  are on the boundary

$$\textcircled{2} g(x) > 0 \text{ inactive: } d_i = 0$$

$$y_i (w^T x_i + b) > 1$$

$x_i$  is beyond margin.



$$w^* = \sum_i d_i y_i x_i$$

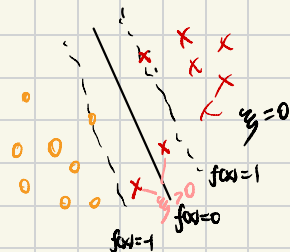
$w^*$  only depends on the points or the margin

$$\sum_i d_i y_i = 0$$

$$(y_i \in \{-1, 1\})$$

## Lecture 9 Support Vector Machine soft margin SVM

soft margin allows some points inside the margin



introduce a slack variable  
 $\xi \geq 0 \quad \forall i$

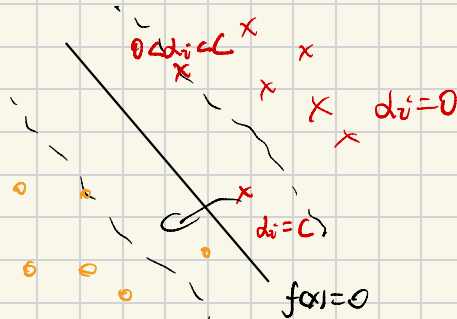
new margin constraint:  $y_i (w^T x_i + b) \geq 1 - \xi_i$   $\xi_i \geq 0$   
 change the allowed value to be inside

new objective:  $\min_w \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i$

s.t.  $y_i (w^T x_i + b) \geq 1 - \xi_i \quad \forall i$   
 $\xi_i \geq 0 \quad \forall i$

Dual SVM:  $\max_{\alpha} \sum d_i - \frac{1}{2} \sum_i \sum_j d_i d_j y_i y_j x_i^T x_j$   
 s.t.  $\sum d_i y_i = 0$   
 $0 \leq d_i \leq C$

$d_i = 0 \rightarrow$  beyond margin  
 $0 < d_i < C \rightarrow$  on the margin  
 $d_i = C \rightarrow$  inside margin



$$w^* = \sum d_i y_i x_i$$

The upper constraint of alpha prevent the outlier points from having a large impact on the final classifier weight